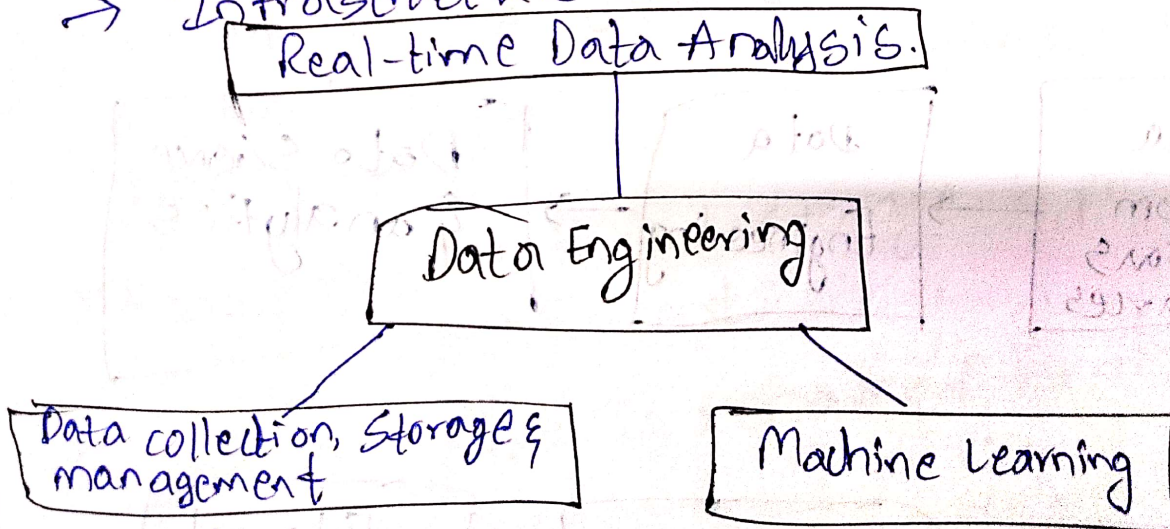


Data Engineering

* It is the practice of designing & building systems for the aggregation, storage & analysis of data at scale.

Role of Data Engineering

- Data Aggregation.
- Data Storage.
- Data Analysis.
- Real Time Insights.
- Strategic Findings.
- Stakeholder Access.
- Data Management.
- Infrastructure Health.



Data Storage

Relational

(Structured)

↓
Ware Houses

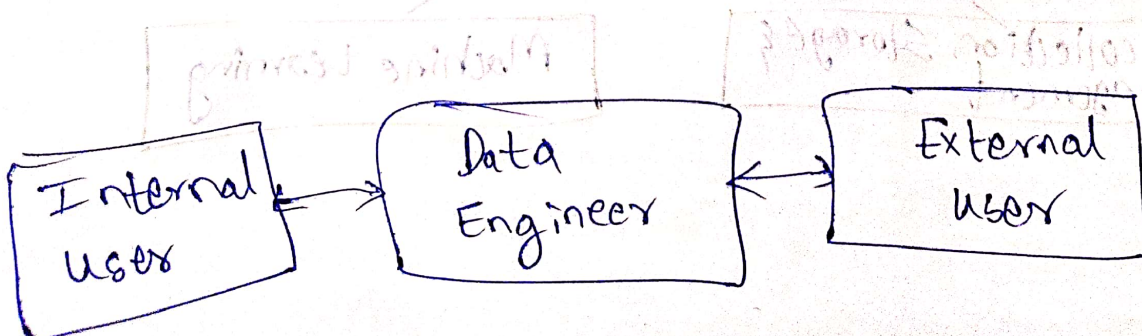
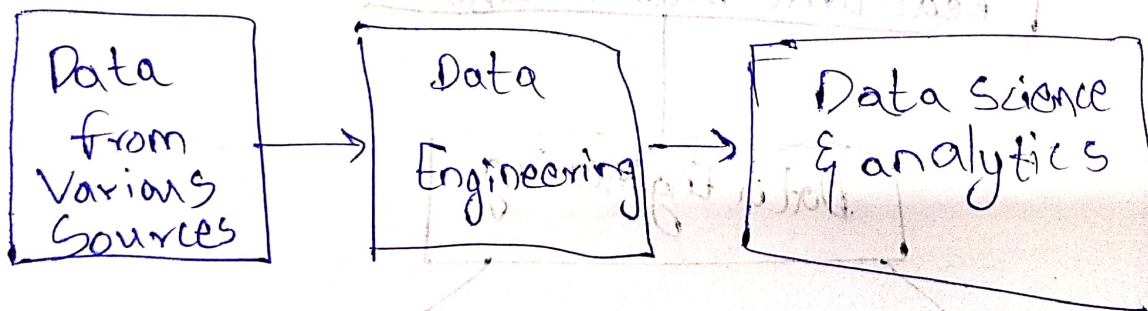
↓
Lake Houses

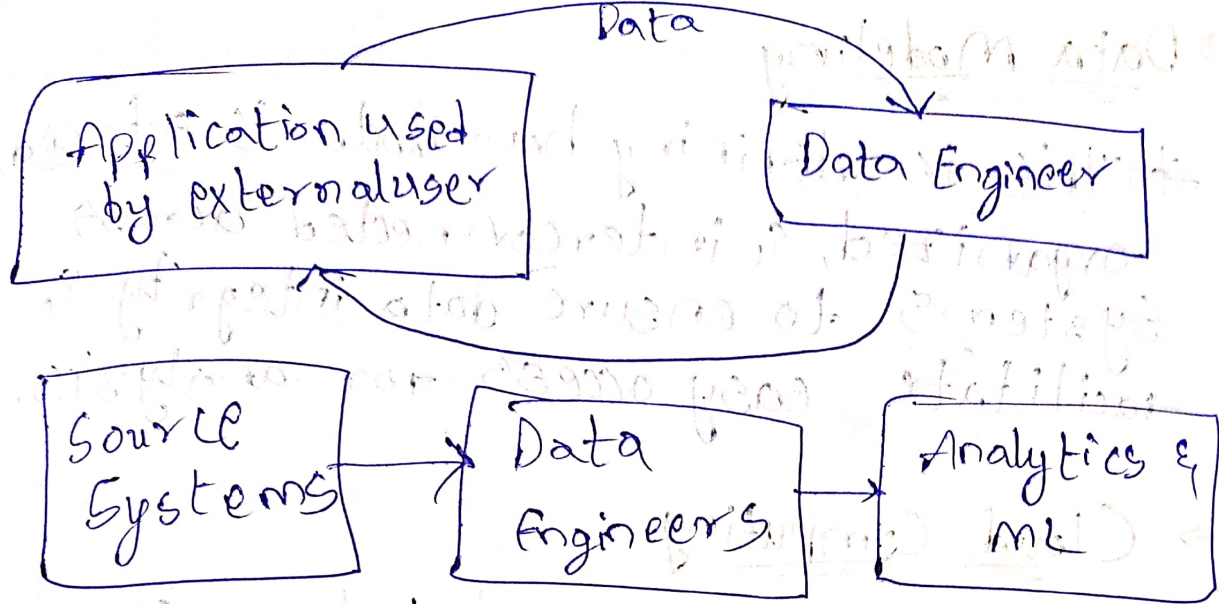
↓
Lakes

↓
NoSQL (Unstructured)

* Data engineering is the intersection of security, data management, DataOps, data architecture, orchestration, & Software Engineering.

* Data Engineers generate essential inputs & data scientists use these inputs for the data analysis.





→ Programming Languages

→ Python is used for data manipulation & data pipeline automation.

→ SQL is essential for managing & retrieving data from relational data bases.

→ Big Data Technologies

→ Apache Spark & Hadoop ⇒ These frameworks enable data engineers to efficiently process large volumes of data quickly.

* Hadoop ⇒ Distributed storage & processing environment.

* Spark ⇒ Spark is renowned for its speed.

* Apache Kafka & Flink are useful for real-time data processing.

→ Data Modeling

* This is defining how data is stored, organized, & interconnected across systems to ensure data integrity & facilitate easy access for analysis.

→ Cloud Computing

* Data systems may depend on cloud services (AWS, Azure, GCP).

→ Data Warehousing

* This allows companies to handle large volumes of organized & unstructured data, guaranteeing data availability for real-time analytics.

Data Engineering Lifecycle

1) Data Generation

* This is where data is created.

→ Sources of Data

- * User activities
- * Applications
- * Sensors & IoT Devices
- * Databases

Data Types

- Structured (tables, rows, columns)
- Semi-structured (JSON, XML)
- Unstructured (text, images, logs).

2) Data Ingestion

* This step collects data from sources & moves it into the system.

Methods =>

→ Batch Ingestion => Data collected at intervals.

→ Real-time Ingestion => Data collected instantly.

Tools => Apache Kafka, Apache Flume.

3) Data Storage

* Data is stored safely for future use.

Storage Types

→ Data Lakes (store raw data)

→ Data Warehouses (store processed data).

→ Databases.

4) Data Processing & Transformation

* Raw data is cleaned & transformed into useful formats.

Processing Types

→ Batch Processing (large datasets)

→ Stream Processing (real-time data).

5) Data Orchestration

* It is the process of automating, scheduling, coordinating & monitoring data work flows.

Data Serving

* Prepared data is made available to users & systems.

- Data Analysis & Visualization
- Monitoring, Quality & Governance.

Modern Platforms for Data Engineering

* Core Components of Modern Data Platforms,

- Data Ingestion (Kafka, Apache NiFi)
- Storage (Data Lakes, Warehouses).
- Processing (Spark, Flink, DBT)
- Orchestration (Airflow, Dagster, Prefect)
- Observability & Governance (Monte Carlo)

Cloud-native Architectures

* Serverless Data Pipelines

* Auto Scaling is present.

* Managed services.

Data Lakehouse

* Merging benefits of data lakes & data warehouses.

Tools ⇒ Delta Lake, Apache Iceberg, Snowflake

Real-time Data Engineering

* Used for fraud Detection, IoT, Personalized Recommendations.

Tools → Apache Kafka, Flink, Spark Streaming.

ML & Analytics Integration

* Feature Stores (Feast & Tecton)

* Data Engineering for ML pipelines.

Ex:- Netflix uses a data platform for personalized recommendations.

* Uses kafka, spark, & Presto.

Def:- Modern data platforms are unified data repositories serves as a foundation for integration & analytics, enabling users with a comprehensive view of their operations, customer behavior, & market trends.

Characteristics

- Scalability
- Real-time processing.
- Flexibility.
- Analytical Capabilities
- Security & Governance
- user-friendly interface
- Seamless integration.

Top 5 Modern Data Platforms

- Sprinkle Data
- Apache Spark
- Google BigQuery
- Amazon Redshift
- Microsoft Azure Synapse Analytics

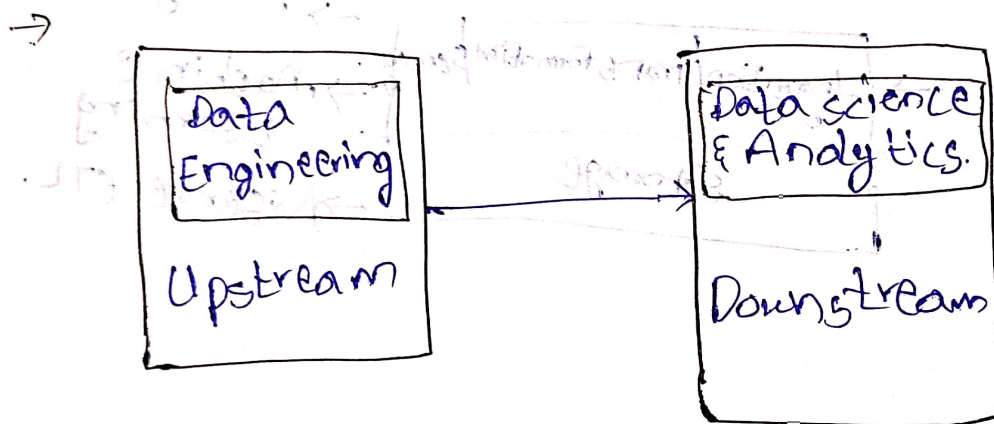
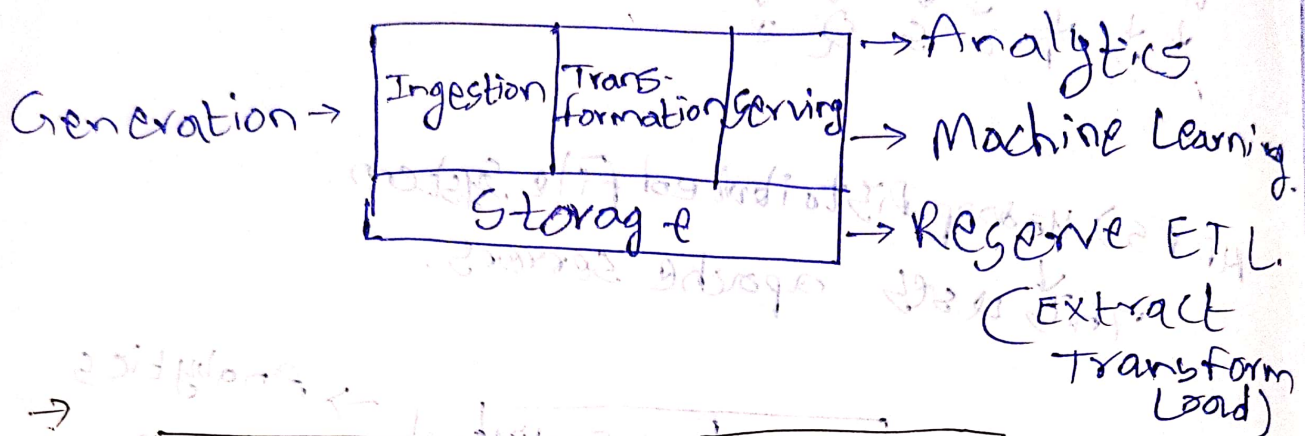
Characteristics

- Scalability
- Real-time processing
- Flexibility
- Multi-cloud capabilities
- Security & compliance
- Integration with existing systems

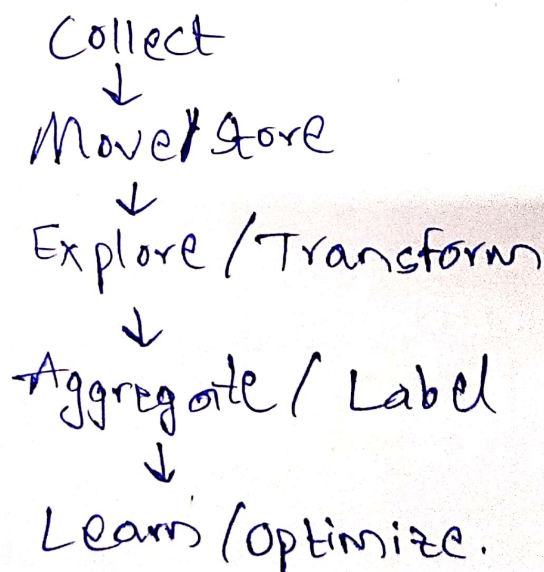
HDFS => Hadoop Distributed File system.

* This uses Apache servers.

Data Engineering Lifecycle



Data Hierarchy



Data Engineering Skills

- Cost
- Agility
- Scalability
- Reuse
- Interoperability.
- Simplicity.

Data maturity

* It is the progression toward higher data utilization, capabilities & integration across the organization.

Starting with data



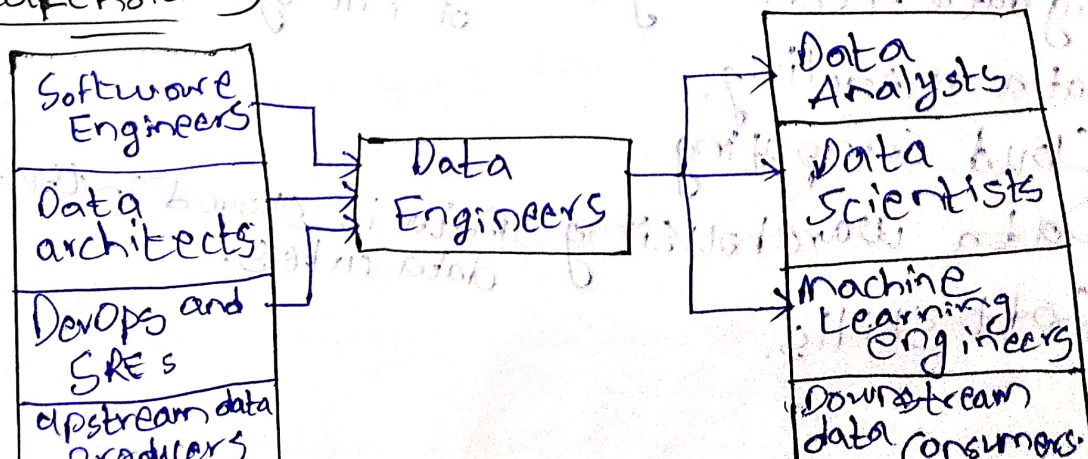
Scaling with data



Leading with data.

Core Responsibilities for Data Professionals

- Effective Communication.
 - Cultural Understanding.
 - Cost Control.
 - Continuous Learning.
 - Achieving outsized value.
- stakeholders



Data Collection

→ Data Collection requires particular attention to compliance & scalability.

Data Delivery

→ Make processed data accessible through secure & performant interfaces.

Tools :-

* GraphQL

* Rest APIs

→ Big Data Technology implements 3V's.

1.) Velocity

2.) Volume

3.) Variety

Metadata :- Data about the data.

Data Engineering Skills Requirement

→ Programming Languages.

* Python, SQL

→ Big data technologies ⇒ Data is stored in the form of file system (HDFS).

→ Data modeling.

→ Cloud Computing.

→ Data warehousing. ⇒ Data is stored in the form of data cubes.

→ Soft Skills.

Data Engineer Roles & Responsibilities

- Data Collection & Integration.
- Data Storage & Management.
- ETL (Extract Transform Load).
- Big Data Technologies.
- NoSQL Databases.
- Cloud Computing.
- Distributed Systems.
- Streaming Data.

Source System

- It is the original place where data is created or generated.
- First stage of data engineering pipeline.

Ex:-

Enterprise Systems

- * These are large-scale applications used by organizations to manage daily business-operations.

Data Sources

- 1) Structured Data.
- 2) Semi-Structured Data.
- 3) Unstructured Data.